

Exp:7-Determination and plotting of Capacitance of isolated and concentric sphere with different inner and outer radius and dielectric constant of the dielectric mediums.

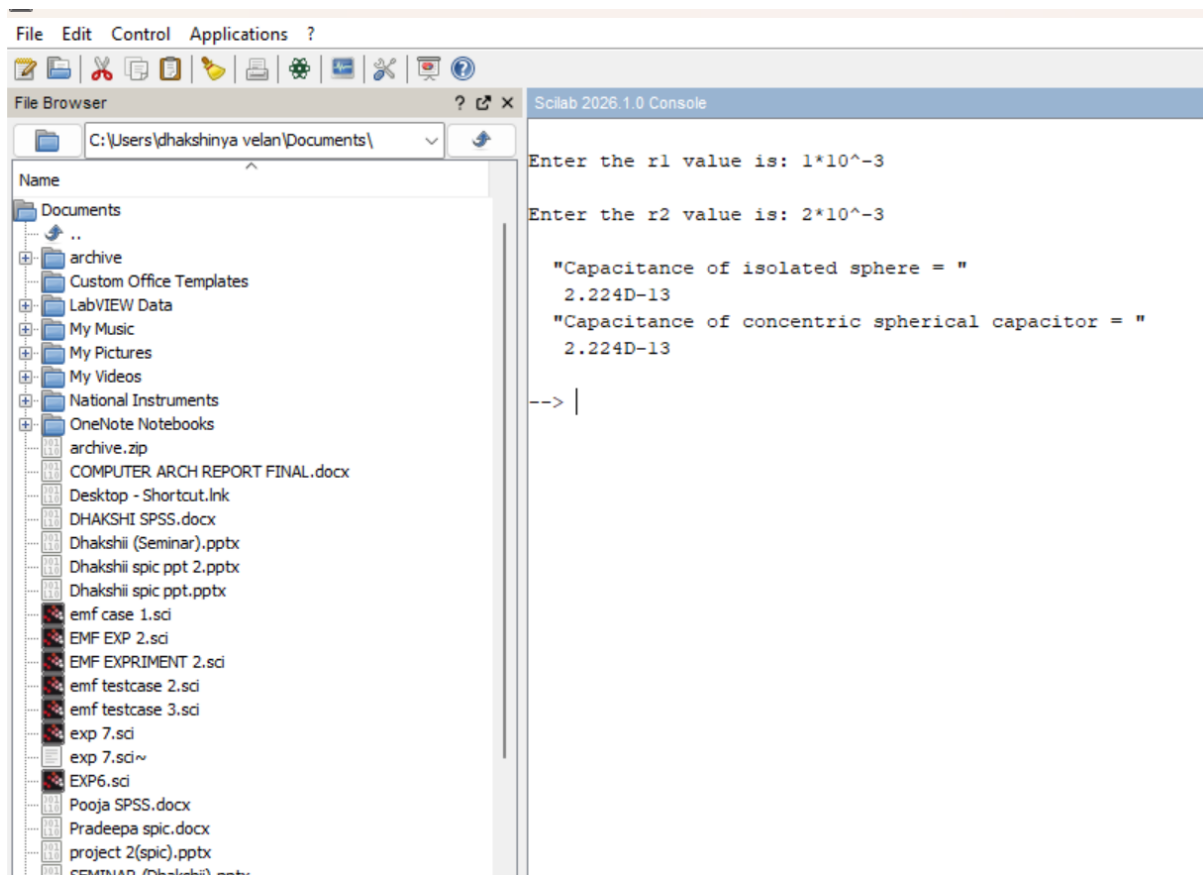
Aim:

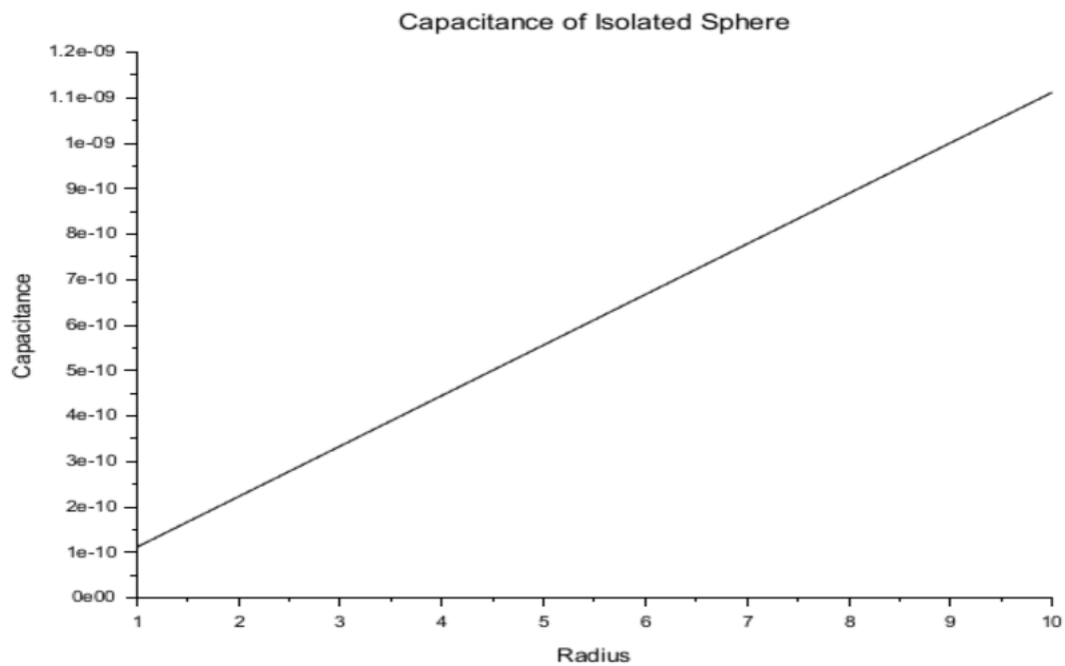
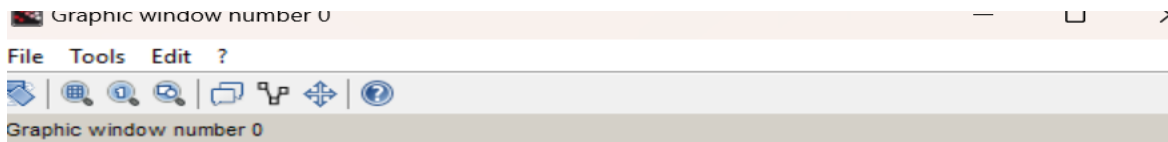
To write program for Capacitance of an isolated sphere and two concentric sphere using SCILAB.

Software required:

- SCILAB version 6.0.1

Partical calucation:





exp 7.sci (C:\Users\dhakshinya velan\Documents\exp 7.sci) - SciNotes

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11 // Capacitance of Isolated Sphere
12 ci = 4 * pi * e * r2;
13
14 disp("Capacitance of isolated sphere = ");
15 disp(ci);
16
17 // Plot
18 r = linspace(1, 10, 50);
19 x = 4 * pi * e * r;
20
21 plot2d(r, x);
22 xlabel("Radius");
23 ylabel("Capacitance");
24 title("Capacitance of Isolated Sphere");
25
26 // Capacitance of concentric spherical capacitor
27 co = (4 * pi * e * r1 * r2) / (r2 - r1);
28
29 disp("Capacitance of concentric spherical capacitor = ");
30 disp(co);

```

Theoretical calculation:

Handwritten calculations for the capacitance of two concentric spheres:

Given:
 $r_1 = 1 \text{ mm}$
 $r_2 = 2 \text{ mm}$
 $\epsilon_r = 3$

Formula for capacitance:
$$C_0 = \frac{4\pi\epsilon_r\epsilon_0}{r_2 - r_1}$$

Calculation steps:
$$C_0 = \frac{4(3.14)(8.854 \times 10^{-12})(2 \times 10^{-3})}{2 \times 10^{-3} - 1 \times 10^{-3}}$$

$$= \frac{2.224 \times 10^{-16}}{1 \times 10^{-3}} = 2.224 \times 10^{-13}$$

Result:

Thus the program was verified for Capacitance of an isolated sphere and two concentric sphere using SCILAB was successfully.